c) 1) $H_0: \beta = 0$
$H_a: \beta \neq 0$

2) $\alpha = .05$

3) $t = \frac{.0034607 - 0}{.0005547} = 6.239$

4) $df = 11 - 2 = 9$

5) $p = .000152$
$p < .05$

6) We reject $H_0$, b/c
$p < \alpha$

d) 1) $H_0: \rho^2 = 0$
$H_a: \rho^2 > 0$

2) $\alpha = .05$

3) $F = 195.5$

4) $df = 1, 82$

5) $p = < 2.2e^{-16}$

6) we reject $H_0$, b/c $p < \alpha$